

AI-Augmented Engineering Case Study for Smart Water Systems

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Abstract

This case study documents four hydrology experiments as an **integrated Smart Water Decision Support Pipeline**, with critical AI collaboration (9 documented corrections, 83% first-pass rate), reproducibility scorecards, manual-vs-AI productivity analysis, and explicit **threats to validity**. It supports the course rubric: AI Collaboration Log (30%), Technical Implementation (30%), Physical Validity (20%), Presentation (20%).

1 Integrated system architecture

Four experiments form one workflow: weather data → rainfall alert → SCS-CN runoff → reservoir optimization → flood inundation mapping, with validation and LaTeX reports at each stage. Optional capstone Streamlit integration wires the same modules.

2 AI collaboration statistics

2.1 Errors caught by experiment

Table 1: Documented AI errors requiring human correction.

Experiment	Errors	Primary types
Exp 1 — Rainfall	3	API misuse, credentials, dry-city logic
Exp 2 — SCS-CN	2	Formula typo, $Q > P$ edge case
Exp 3 — Reservoir	2	Optimizer infeasibility, trade-off bounds
Exp 4 — Flood	2	Depth convention, monotonicity
Total	9	All logged in <code>prompt_log.md</code>

2.2 Prompt / output summary

Table 2: Critical AI use metrics (Experiments 1–4).

Metric	Count
Major AI deliverables reviewed	52
Accepted without material correction	43
Required human correction	9
Effective first-pass rate	83%

3 Manual vs AI-assisted benchmark

Author estimates for comparable scope (single developer): AI roughly halved implementation time;

Smart Water Decision Support Pipeline

Four specialized experiments as one integrated hydrology workflow

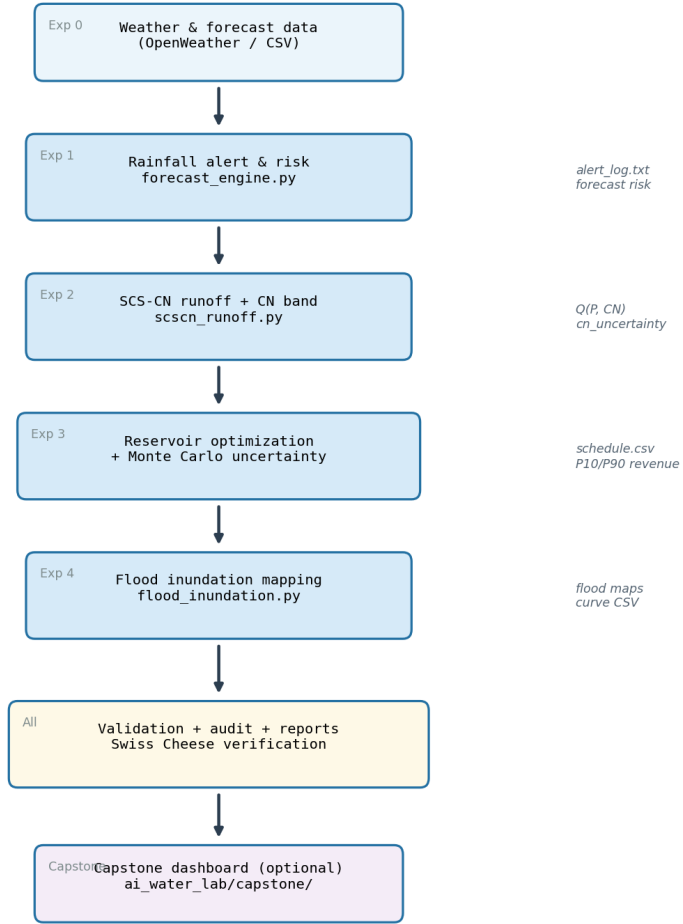


Figure 1: Smart Water Decision Support Pipeline (Experiments 1–4).

Table 3: Productivity comparison (hours).

Task	Manual	AI-assisted	Verification
Exp 1 — API + forecast	4	1.75	1.0
Exp 2 — SCS-CN + CN band	6	2.5	0.75
Exp 3 — Optimization + MC	8	4.0	1.5
Exp 4 — Flood + analysis	6	2.5	1.0
Reports + case study	12	7.0	cross-read
Total	36	18	~5

verification remained $\sim 25\text{--}30\%$ of effort (Jagged Frontier).

4 Reproducibility scorecards

Table 4: Per-experiment reproducibility (= present).

Item	Exp 1	Exp 2	Exp 3	Exp 4
README				
AGENTS.md				
prompt_log.md				
pytest	20	23	23	22
Validation CLI				
Auto figures				
requirements.txt				
audit_summary.txt				
LaTeX report				

5 Threats to validity

- **OpenWeather data:** sparse **rain** fields; mitigated by demo mode and 50-city batch.
- **SCS-CN assumptions:** lumped CN; mitigated by uncertainty band $CN \in [75, 85]$.
- **Reservoir inflows:** simplified 7-day vector; mitigated by Monte Carlo and drought infeasibility.
- **Flood model:** bathtub approximation; documented limitations and seed sensitivity.
- **LLM outputs:** hallucination risk; Swiss Cheese verification; 9 corrections documented.

6 Swiss Cheese verification

Five layers per experiment: (1) AI output review, (2) `pytest -v` (88 total), (3) physical rules, (4) validation CLI, (5) LaTeX + terminal evidence.

7 Engineering maturity

8 AI outputs challenged (summary)

Nine corrections including: missing rain $\rightarrow 0$ mm/h; env-based API keys; correct SCS- S formula; feasible reservoir x_0 ; honest drought infeasibility; non-negative flood depth; monotonic stage curves. Full table in `AI_Engineering_Portfolio.md`.

9 Deliverables map

- Code: `ai_water_lab/experiment{1,2,3,4}...`
- Pipeline: `ai_water_lab/SMART_WATER_PIPELINE.md`
- Reports: `Experiment-reports/Experiment{1,2,3,4}...`
- Capstone: `ai_water_lab/capstone/`

Submission checklist

- Four experiment PDFs + this case study PDF
- Figures `smart_water_pipeline.png`, `case_study_radar.png`
- All 88 pytest cases green
- Cross-check with ChatGPT, DeepSeek, Gemini

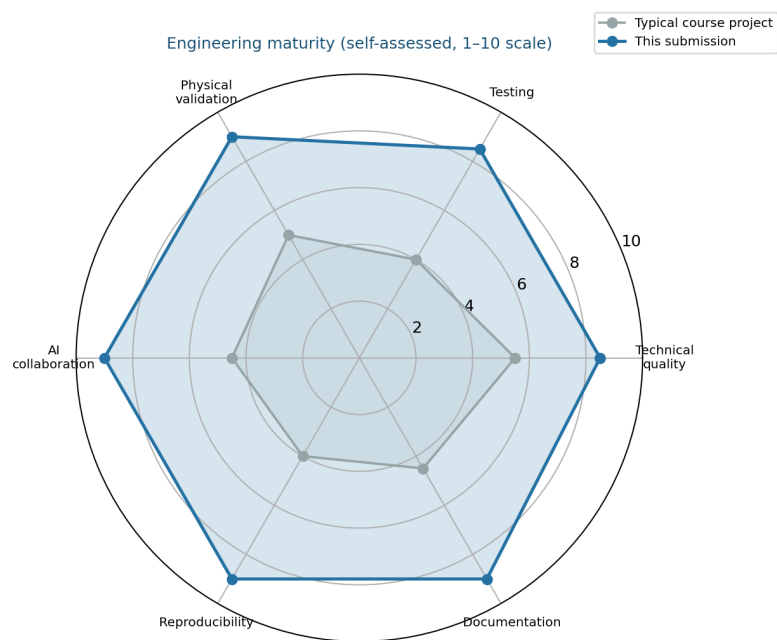


Figure 2: Self-assessed maturity (1–10): typical course project vs this submission.